



Final App Concept (Upgraded)



“Smart Task Manager Pro” (Full-featured version)

This is NOT just a CRUD app anymore — it becomes a **mini productivity system**.



FULL FEATURE SET (Optimized for Your Assignment)



1. Authentication (Required)

- Register + Login
- Store user session (SharedPreferences)
- Logout feature (BONUS ☒)

👉 Use:

- MockAPI (recommended)
OR
 - Local storage (fallback)
-



2. Task Management (CRUD – Core Feature)

Task Fields:

- ID
- Title
- Description
- Category (Work, Study, Personal)
- Priority (Low, Medium, High)
- Status (Pending, In Progress, Completed)
- Created Date

👉 Screens:

- Add Task
- Edit Task
- Delete Task
- View Task List

3. Search

- Search by **task title**
- Real-time filtering while typing

👉 Example:
Typing “meet” → shows “Meeting with client”

4. Advanced Filtering (VERY IMPORTANT ★)

Add a filter panel:

- By Category
- By Status
- By Priority

👉 You can combine filters:
“High Priority + Pending + Work”

💡 UI Tip:
Use **Chips** / **Dropdowns** / **Bottom sheet**

5. API Integration (HIGH PRIORITY ⚠)

Use:

- Retrofit (Java)

Endpoints:

```
/users  
/tasks
```

👉 This is where most candidates fail — DO IT PROPERLY.

6. Offline Support (BIG BONUS ★)

Use:

- SQLite (or Room if you want advanced)

Flow:

- Load from API → cache in DB
- If no internet → load from local DB

👉 This will impress them A LOT.

7. Dark / Light Mode (BONUS)

- Use AppCompatActivity theme switching
 - Save preference
-

8. Error Handling (VERY IMPORTANT)

Handle:

- No internet
- API failure
- Empty list
- Invalid input

👉 Example:

- Show Toast / Snackbar
 - Show “No tasks found” UI
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9. User Info Display (BONUS)

- Show logged-in username on Home screen
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ARCHITECTURE (THIS IS YOUR DIFFERENTIATOR)

Since you're using Java:

Use: MVVM (simplified version)

Structure:

```
com.yourapp
├── activities
├── adapters
├── api
├── database
├── models
├── repository
├── viewmodel
└── utils
```

Flow:

```
UI (Activity)
↓
ViewModel
↓
Repository
↓
API + SQLite
```



UI STRUCTURE (XML Based)

Screens:

1. Splash Screen
 2. Login Screen
 3. Register Screen
 4. Home (Task List)
 5. Add/Edit Task
 6. Task Details
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Home Screen Layout Idea:

- Top: Search bar
- Below: Filter options
- List: RecyclerView
- Bottom: Floating Action Button (+)



TECH STACK (JAVA VERSION)

- Language: Java
- UI: XML + RecyclerView
- API: Retrofit
- JSON: Gson
- Local DB: SQLite / Room
- Image/Icon: Material Icons



HOW TO MAKE IT LOOK “PRO LEVEL”



Use:

- CardView for tasks
- Proper spacing & margins
- Consistent colors
- Loading indicator (ProgressBar)



EXTRA FEATURES (OPTIONAL BUT POWERFUL)

If you still have time:

- Swipe to delete (RecyclerView)
- Sort tasks (by date/priority)
- Pull-to-refresh
- Animations (basic)



INTERVIEW EXPLANATION (VERY IMPORTANT)

When they ask:

👉 “Explain your app”

You say:

“I built a Task Management Android app using Java with MVVM architecture. It supports authentication, full CRUD operations, search, and multi-criteria filtering. I integrated REST APIs using Retrofit and implemented offline support using SQLite. The app also includes proper error handling, dark mode, and a clean, user-friendly UI.”

🌟 That answer alone can impress.



FINAL ADVICE (CRITICAL)

Don't:

- Overcomplicate UI
- Skip API integration
- Mix all logic inside Activities

Do:

- Keep code clean
- Separate layers
- Test all flows